

Final Project of MENG 2020 – FALL2025

This project can be done individually (not recommended – need permission from your professor), or in a team of two (ideal).

1. Introduction:

On this project you will make calculations and conclusions on patients' muscle movement indexes (called psoas major muscle index or PMMI in short) for patients in their 40s, who got treatments under a special care unit, at specific Canadian hospitals in four different provinces and two international hospitals. This data is produced using one specific sensor and the unit is not important.

A file contains readings of information of patients' muscle movement collected at different hospitals. You are to produce a detailed report on this data.

2. Procedures:

1. Read the data in the excel worksheet file PMMI-2025.xls into a matrix to start.
2. Calculate the average of patients' muscle movement readings for all the hospitals in the file (one single average calculated for all hospitals in all months).
3. Calculate the average patients' muscle movement for each hospital for the entire year (one average per hospital). Print the results in one column.
4. Calculate the average patients' muscle movement for each province that has a hospital in the date file for the entire year (one average per province/each area outside Canada). Print the results in one column.
5. Calculate a month-by-month average of patients' muscle movement of each province and draw it on a graph (60 averages – 12 months for 4 provinces and for both hospitals outside Canada). Print the results in the matrix format.
6. Calculate a month-by-month average of patients' muscle movement of all hospitals and draw it in one graph (12 averages – one per month).
7. Draw a graph combining individual data for all hospitals in all months (graph from raw data – choose a different color or marker for each hospital).
8. Pick up the 2 hospitals' data from Toronto and plot it in 2 sub-plots in a year month-by-month.
9. Calculate and draw the average patients' muscle movement **categories** for all hospitals in each month. Color your graph based on the following categories and color scheme:

Patients' Muscle Movement Index Categories and Color Scheme

(PMMI)	Category	color
0-15	Very Good	
16-31	Good	

Final Project of MENG 2020 – FALL2025

32-49	Moderate	
50-99	Poor	
100+	Very Poor	

10. Can you infer some fit (trend) between the months and the patients' muscle movement indexes? Is it the same/similar fit for all the hospitals? Write your conclusions in the report.
11. Compare the data in Montreal and Vancouver with the **moderate level** PMMI and show which one contains more **moderate level** movements. Print your conclusion on the screen.
12. Compare the data in the two International hospitals and Vancouver with **the poor level** PMMI, and show which one contains **more poor level** movements. Print your conclusion on the screen.
13. Calculate the average of **moderate level** patients' muscle movement for each province (or international area) in the entire year (one average per province/area). Print out your results in a column.
14. Calculate the average of **very good level** patients' muscle movement for each hospital for the entire year (one average per hospital). Print out your results in a column.
15. Calculate and plot month-by-month averages of **very good level** patients' muscle movement of all hospitals on one graph (12 averages – one per month).
16. Calculate and plot month-by-month average of **good level** patients' muscle movement of each province with a graph (60 averages – 12 months for 4 provinces and both hospitals outside Canada).

3. Notes:

1. All computations and graphs are to be done with MATLAB.
2. You are to write a report. Your report must have an introduction about the purpose of the report and **explanation and responsibilities (what did each team member contribute to the project?) of each team member. Each team needs to submit one report.** Your report must have a cover page (that clearly shows group name and your names). messy, and unorganized report will be not accepted.
3. The report must be detailed, well presented and attractive. Don't be afraid to use colors to emphasize parts of the report (for example when the patient's muscle movement is poor or very poor). Be creative in the use of tables, graphs and images. Points will be awarded to the exactness of the computations, appearance, ease of reading, and the quality of the English language. Be original! Plagiarism will be dealt with severely and get zero marks immediately.
4. Your report must have a conclusion, especially about point 10 above in the procedures. You must also report about your experience doing this project and how you would do things differently if you had to do this again.
5. Your report must have a cover page clearly identifying all teammates and the date.

6. Finally, as an appendix, your report must contain *all source code* of your MATLAB code. The code must have comments to explain the procedures you did and the idea of your statements. Your report must be submitted based on professor's provided instructions. Be original! Plagiarism will be dealt with severely to the full extent of Humber Polytechnic academic integrity regulations. The Turnitin system will be used to help the markers in their assessment of originality. Also, a reminder that, you may use Generative AI (e.g. ChatGPT, Grammarly, Perplexity, DeepL Translator) for grammar correction only, not doing the project, or codes. Failure to stay within these limits will be considered a breach of academic Policy.

4. Submission Requirements:

1. Team members must be determined and reported to your professor by November 19, 2025(Consent form will be at the project folder, and should be submitted by this due date).
2. Teammates must be clearly identified as per point #2 in the notes above.
3. Your report will be one single word file. All codes should be added to the same word file. This is the same as your lab report. Please do not attach Pdf, or other formats.
4. Project must be submitted on or before due date, as a final project, **late submissions won't be accepted (November 30.2029)**. If you are concerned about getting the assignment in on time, submit it early (Do not leave the submission for the last day, any issue might happen). Technical excuses will not be accepted. Team members should work equally, otherwise, the mark will not be given. It is students' responsibility to manage the work and the distribution off all the tasks. **Only one submission from a team is needed.**